Graduate Project Presentation

Computer Vision Project

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Contents

- **Introduction**
- Data cleaning and sampling
- Exploratory data analysis
- Disaster Type Classification
- Damage Level Classification
- Conclusion

Introduction

- Mission:
  - Support crisis response agencies by automating building damage assessment using satellite imagery.
  - Reduce reliance on slow manual annotation during disasters.
- Dataset:
  - Derived from the xView2 Challenge Dataset
  - Disaster types (3):
    - Midwest Flooding
    - SoCal Fire
    - Hurricane Matthew
  - Damage levels (4):
    - 0-3: no-damage -> destroyed

Introduction

- Project Components:
  - Exploratory Data Analysis (EDA)
    - Dataset statistics & label distribution
    - Feature exploration: color channels, edge information
    - Evaluate whether color features distinguish
  - Modeling Task A: Disaster type classification
    - Predict flooding vs. fire
  - Modeling Task B: Damage level classification
    - Predict hurricane building damage (0–3)

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Data Sampling and Cleaning

- **Data Sampling**
  - Total data: 200 images for each disaster
  - Shuffling: np.random.default_rng(seed=42)
  - Splitting: 80/20 split, 480 images for training, 120 images for testing
- **Data Cleaning**
  - Total data: 200 images for each disaster
  - Valid data: 200 images for each disaster
  - No invalid/missing data, no NAN value

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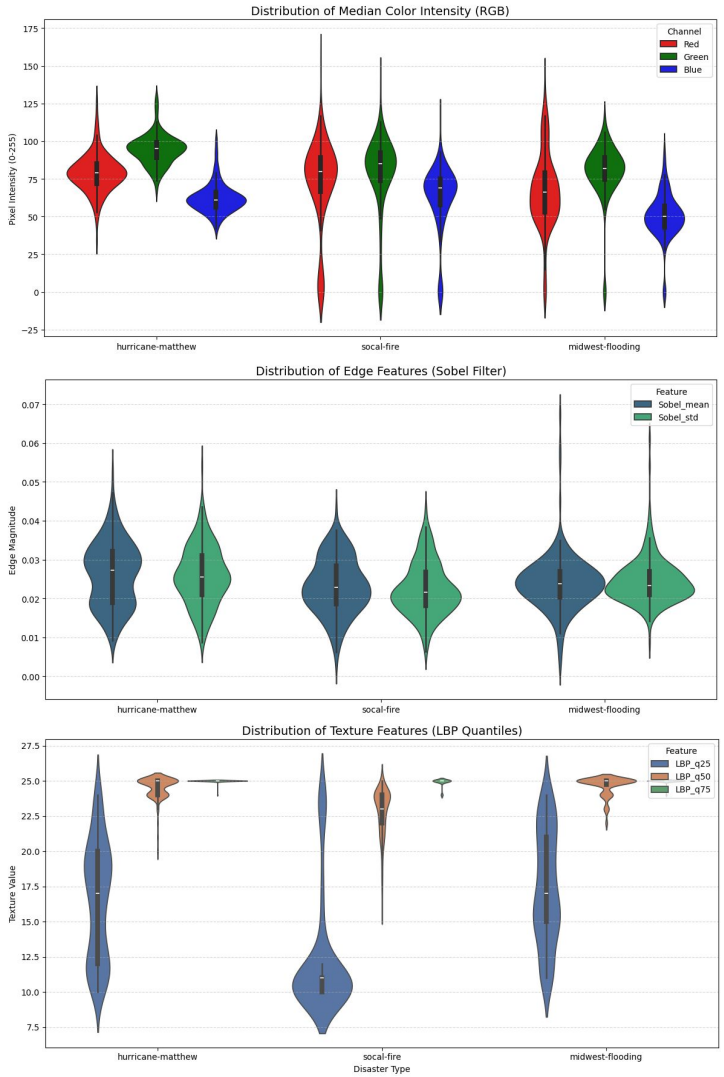
Disaster Type

- Color:
  - SoCal Fire: higher red values
  - Midwest Flooding: higher blue/green
  - Hurricane Matthew: moderate green

Color strongly separates flood vs. fire.

- Edges & Texture:
  - Fire: stronger, irregular edges; more uniform textures
  - Flood: smooth surfaces with low edge intensity
  - Hurricane: intermediate patterns

Structural features further support classification.



Damage Level

- Color:
  - Low damage: stable RGB values
  - Severe damage: broader intensity ranges

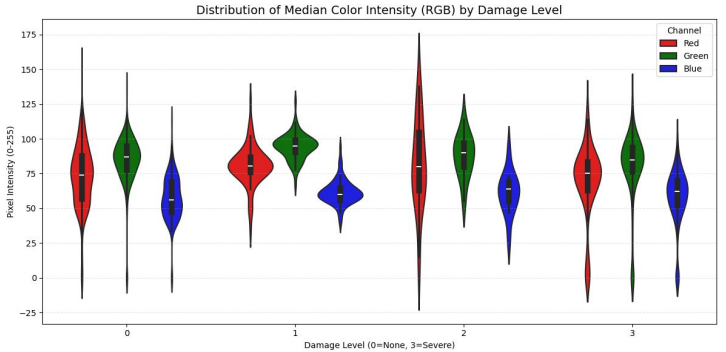
Color helps, but overlaps across classes.

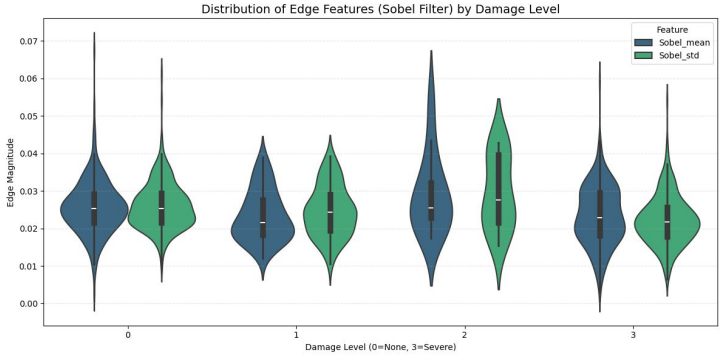
- Edges:
  - Low damage: smooth edges
  - Severe damage: higher edge magnitude + variance

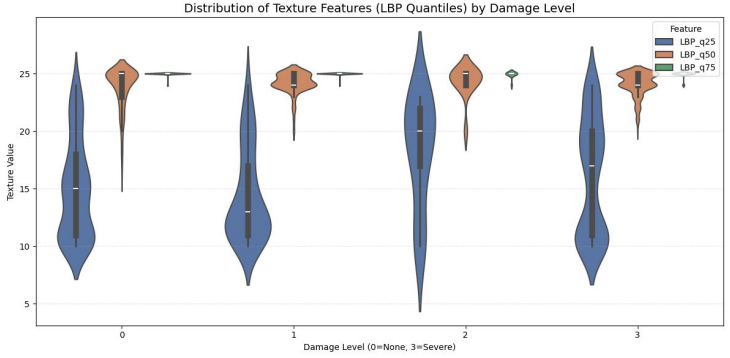
Strong indicator of structural breakage.

- Texture:
  - Low damage → regular textures
  - High damage → irregular, high-variance LBP

Reinforces edge-based separation.







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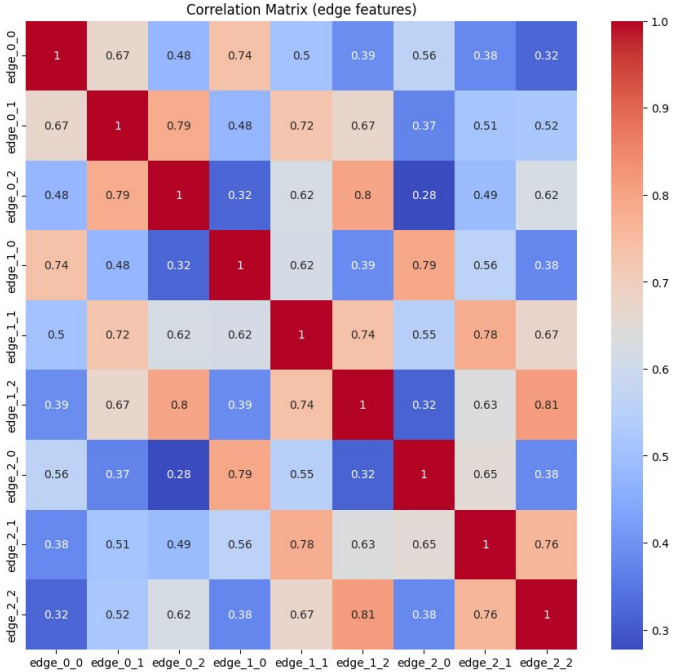
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- **Disaster Type Classification**
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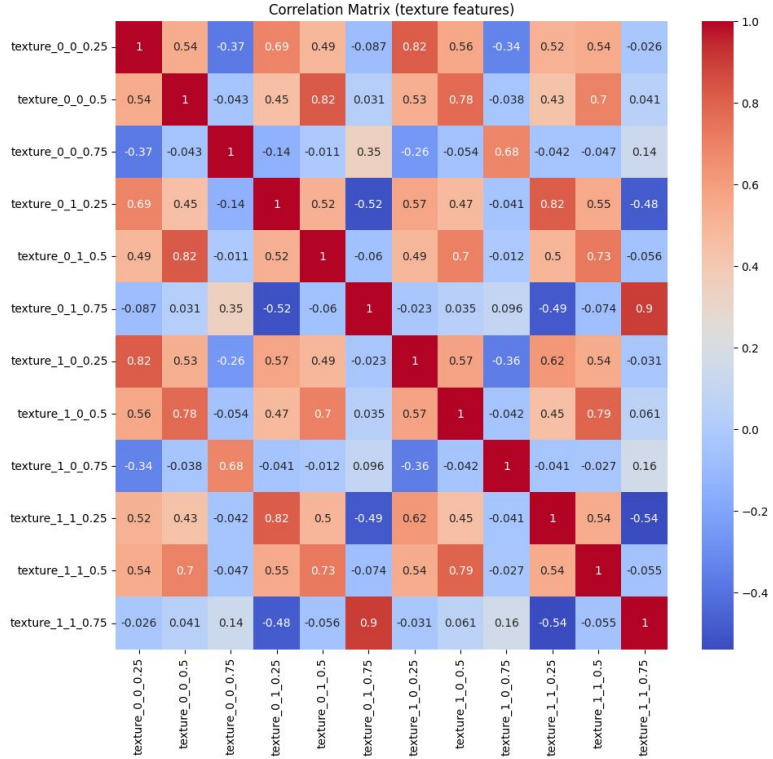
Task 1: Disaster Type Classification

- Model: Logistic Regression (L2 Regularization) → hyperparameter
- Feature Engineering:
  - Color (3):
    - Calculated mean values of RGB channels.
  - Edge (9):
    - Image divided into **3×3 blocks** to capture localized terrain and topography.
    - Calculated average edge density for each block.
  - Texture (12):
    - Image divided into **2×2 blocks** to retain local texture information.
    - Extracted **Q1, Median, and Q3** pixel values from the LBP-processed image for each block.

Task 1: Disaster Type Classification

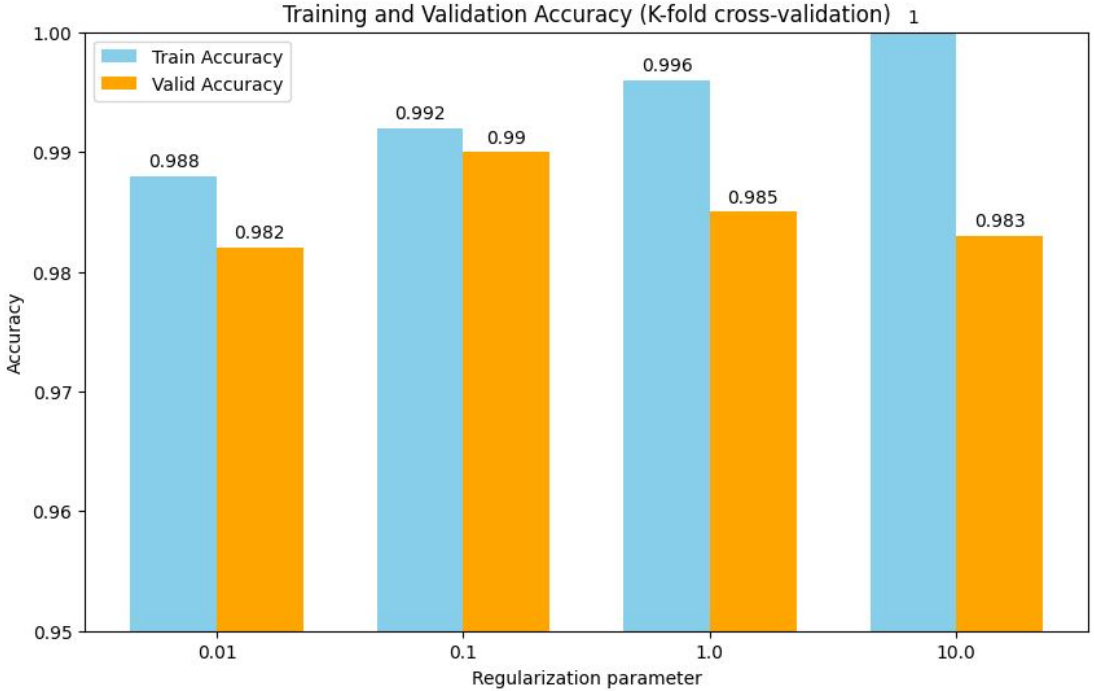
☆ Check the correlation among localized features:





Task A: Disaster Type Classification

- K-fold cross validation (K=5) for regularization coefficient λ



- When **λ = 0.1**, the model has highest validation accuracy. → best choice
- The average accuracy achieves **0.990** in validation set.

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Feature Engineering

Color (9) :

- Quartiles of the global color of the images

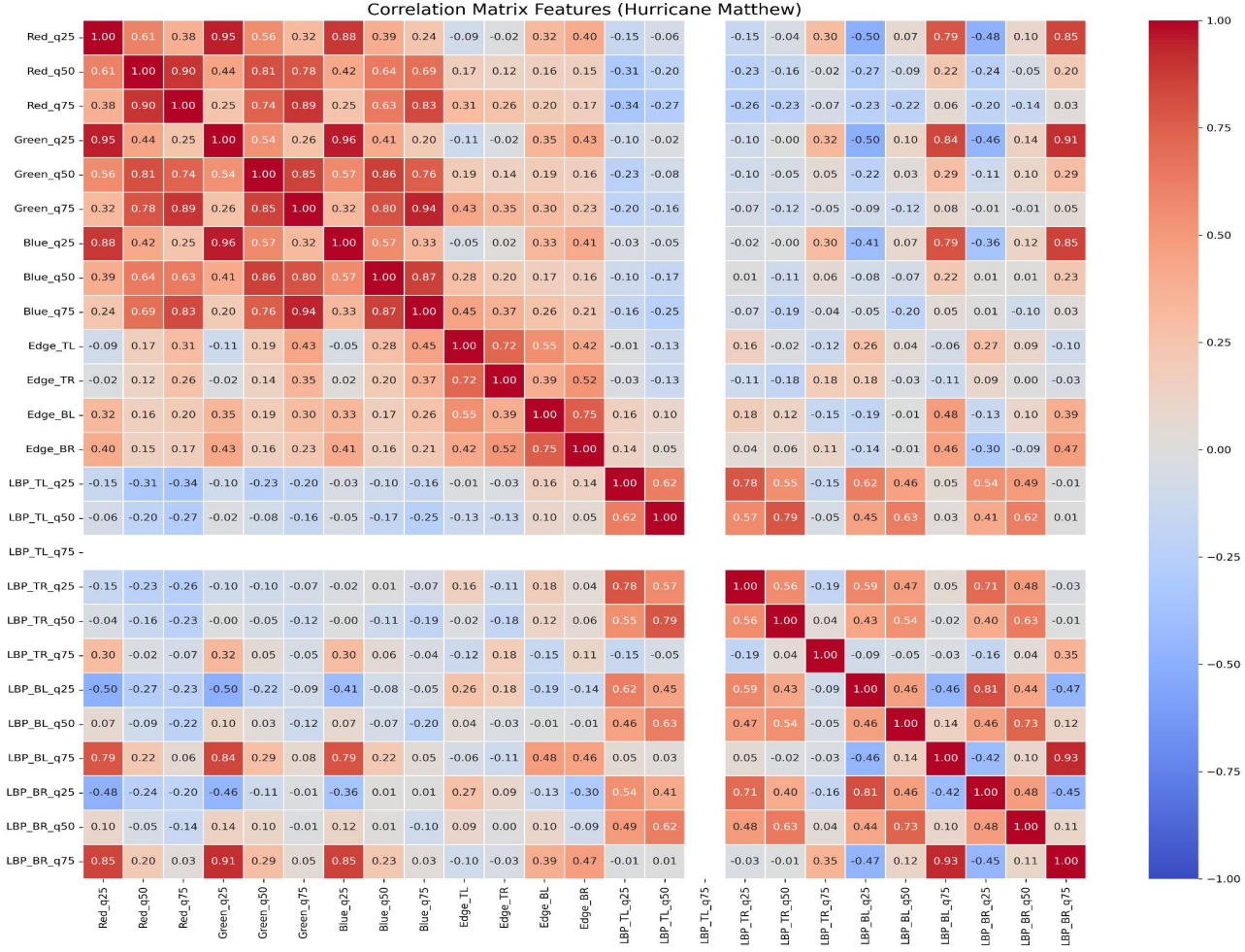
Edge (4):

- Image divided into 2×2 blocks to capture localized terrain and topography.
- Calculated average edge density for each block.

Texture (12):

- Image divided into 2×2 blocks to retain local texture information.
- Extracted Q1, Median, and Q3 pixel values from the LBP-processed image for each block.

Feature engineering : correlation matrix



Modeling Strategy & Optimization

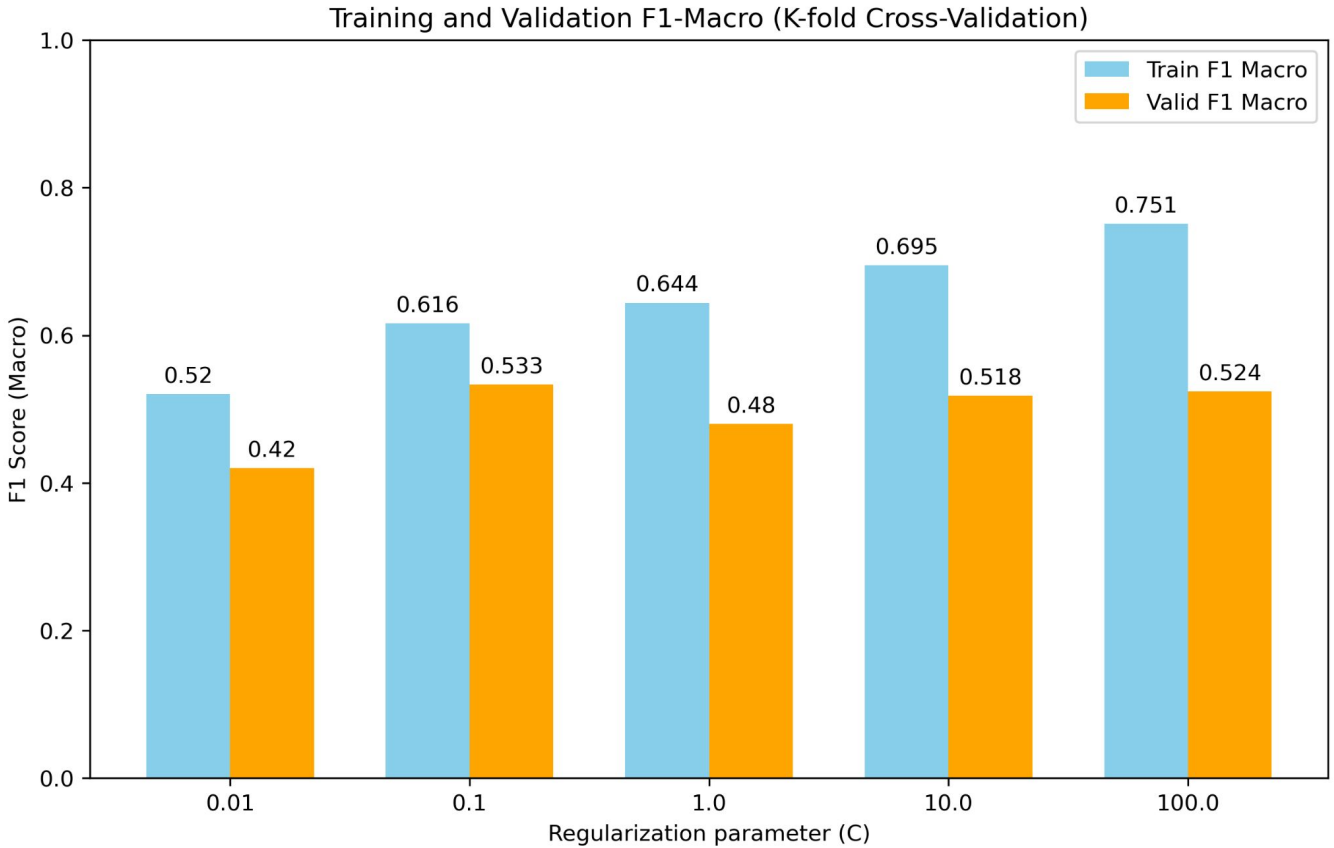
➔ Algorithm Selection

- Explored Random Forest (for non-linearity) and Logistic Ordinal Regression (for interpretability and weights).
- **Final choice**: Weighted Logistic Regression with StandardScaler for convergence.

➔ Addressing Imbalance :

- Classes 1 and 2 (Minor/Major damage) had extremely low support (< 10 images).
  - **Solution 1:** Class Weights: Assigned heavy penalties (10x) for errors on rare classes.
  - **Solution 2:** Data Augmentation: Synthetic generation (flips/rotations) applied strictly to the training set to increase minority class exposure.

Model Validation (K-Fold Cross-Validation)



Final weighted F1 score on test set : **0.55**

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Takeaways

- **Task A**: Logistic Regression (with L2 Regularization)
    - K-fold cross validation -> Validation score: 0.9900
- **Task B**: Ordinal Logistic Regression
    - K-fold cross validation -> Validation F1 weighted score : 0.55

Strengths and limitations

- Strengths:
  - We split the image into blocks to extract “local” features, increasing the spatial resolution.
- Limitations:
  - For task B, the score are still low for a real world application context.

Thank you!

Computer Vision Project

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